AUTONOMOUS EMERGENCY RESPONSE PLATFORM

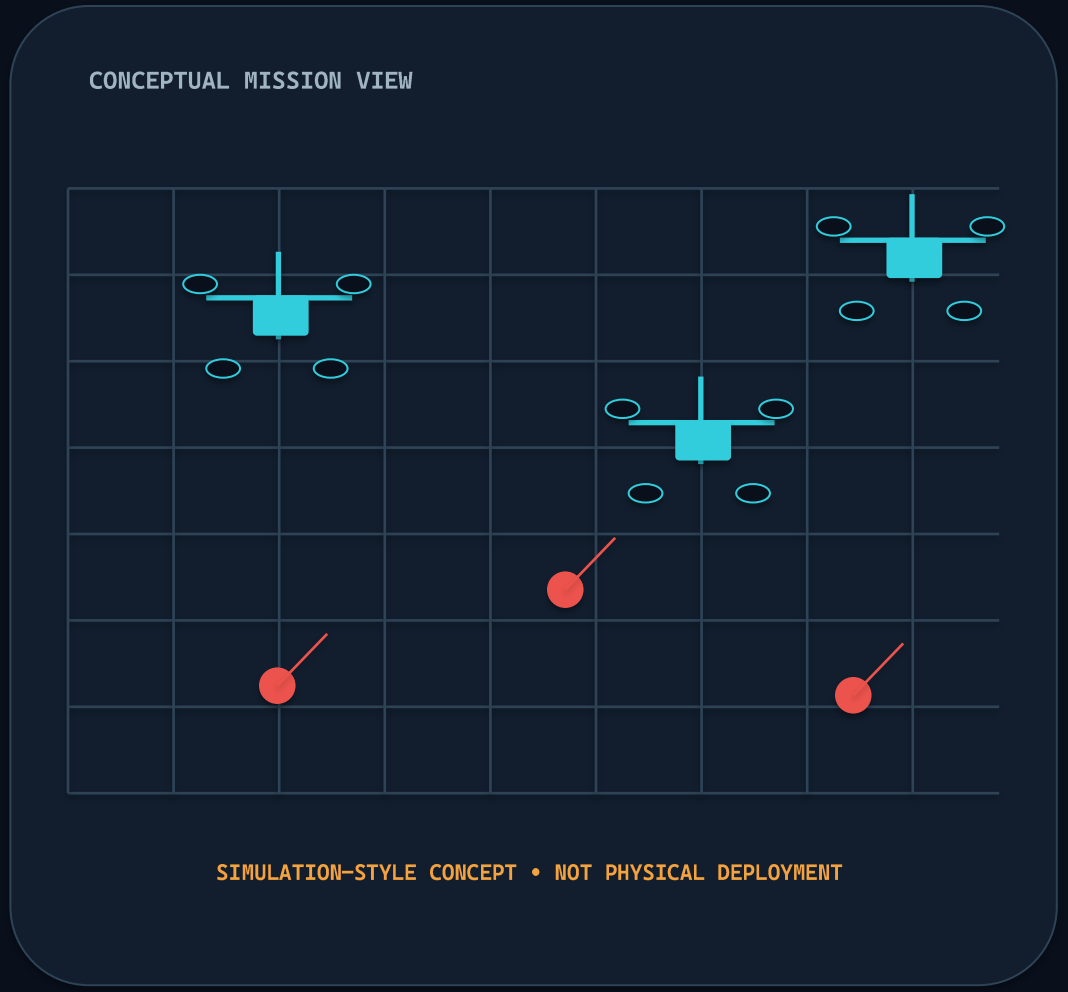
RescueSwarm

Autonomous Multi-Drone Search, Survivor Geolocation & Kit Delivery

SEARCH • DETECT • LOCATE • DELIVER • RETURN

One operator. Multiple autonomous vehicles.
One coordinated rescue mission.

AUTONOMOUS | COLLABORATIVE | OFFLINE | EDGE AI



PROBLEM

The Rescue Challenge

Detection alone is not enough.

A rescue system must detect, locate, coordinate and respond under time pressure.

CONVENTIONAL SEARCH

Large areas • manual search
Approximate visual sightings
Aid arrives after the search

THE BOTTLENECK

One vehicle creates a coverage and response constraint.
Connectivity may be unavailable.

THE REQUIREMENT

Parallel search
Reliable survivor coordinates
Aid delivery before recovery

THE OPPORTUNITY / TURN A FLEET INTO A COORDINATED RESCUE SYSTEM

SOLUTION

RescueSwarm

From mission boundary to safely completed response.



One operator supervises. The software executes the mission.

AUTONOMY

After START, mission logic drives the fleet.

COORDINATION

Shared targets, ownership and reallocation.

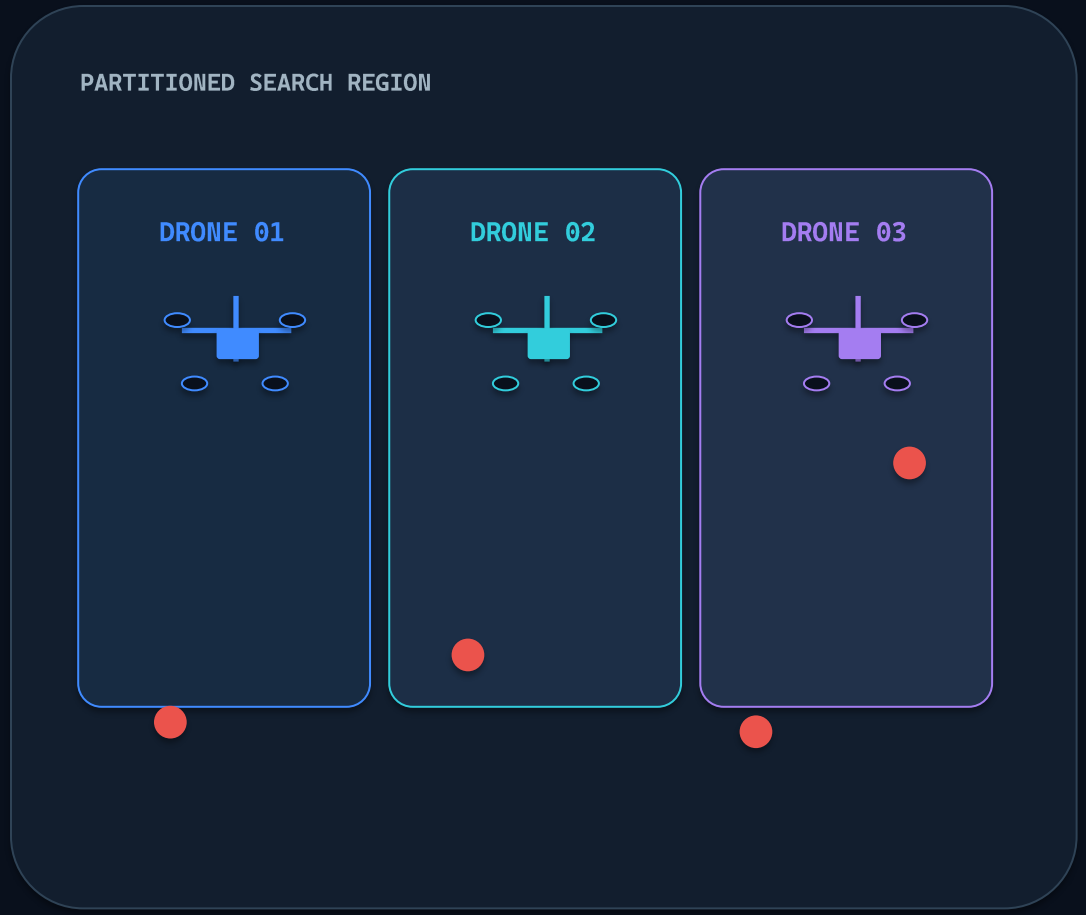
SAFETY

Native RTL/LAND, geofence and recall paths.

Designed Around the Full Rescue Mission

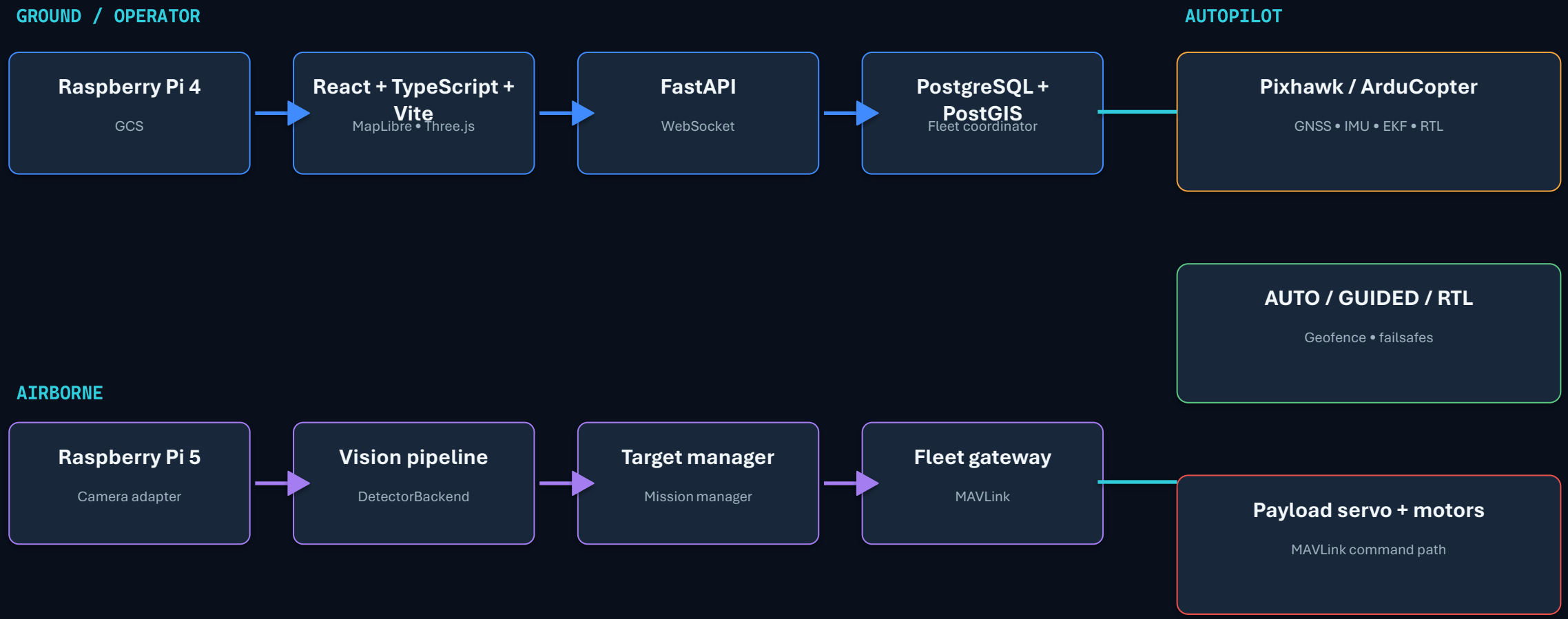
A software-scale rehearsal of the operating envelope.

- Search area up to approximately 10 hectares
- Up to 10 survivors
- 2+ collaborative autonomous drones
- Local / offline communication
- One operator / one GCS
- Autonomous operation after START



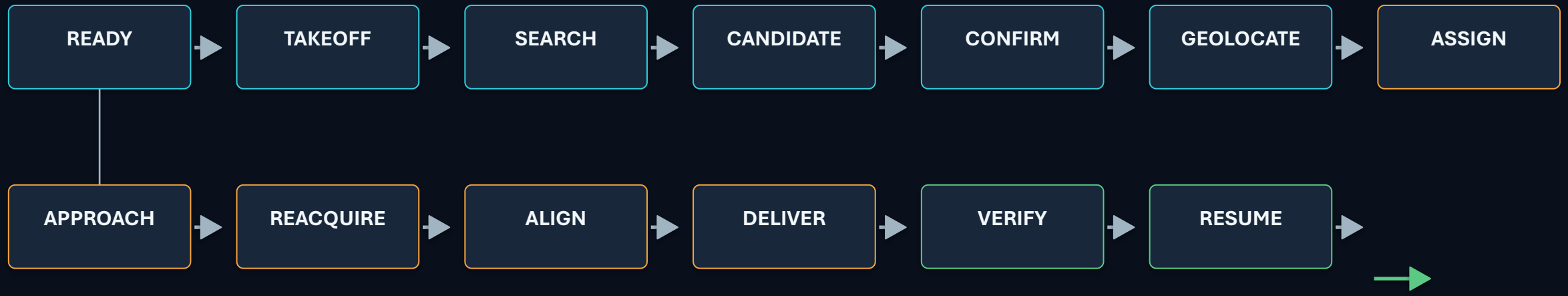
DESIGNED TARGET • NOT CLAIMED AS COMPLETED PHYSICAL TESTING

End-to-End System Architecture



LOCAL NETWORK ONLY • MAVLink • NO CLOUD DEPENDENCY

Autonomous Rescue State Machine



MISSION COMPLETE | RTL | LAND | DISARM

SAFETY PATHS: BATTERY • LINK • EKF • GEOFENCE • EMERGENCY RECALL → RTL / LAND

Fleet Intelligence

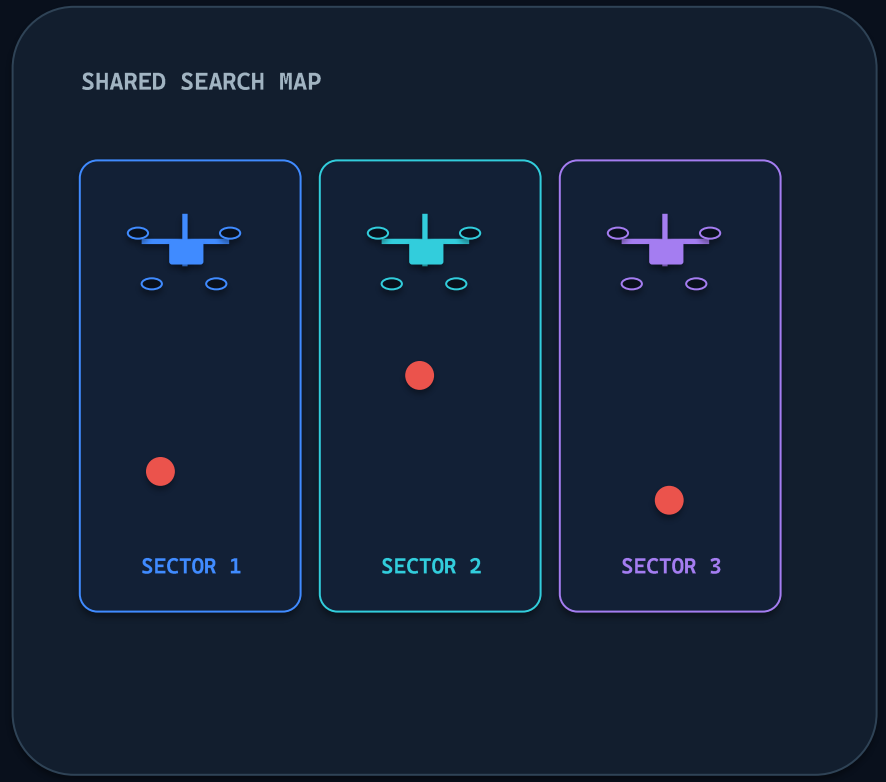
Central coordination. Independent vehicle agents.

FLEETMISSIONCOORDINATOR

Deterministic partitioning
Shared target registry
Duplicate merge
Transactional assignment
Emergency recall

VEHICLERESCUEAGENT

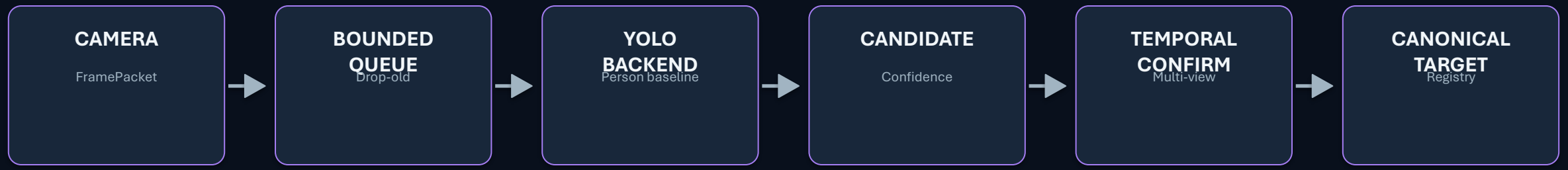
Independent autonomy
Own mission state
Own health and link
Delivery ownership
Safe recovery paths



VEHICLE LOSS → REMAINING-AREA REALLOCATION → NO DUPLICATE DELIVERY

VISION

Pixel-to-Survivor Pipeline



ENGINEERING GUARDRAIL

Bounded queues prevent unlimited frame backlog.

MODEL POSTURE

Generic person detector is current baseline.

HARDWARE PATH

Monochrome can replicate to 3 channels for RGB-shaped models.

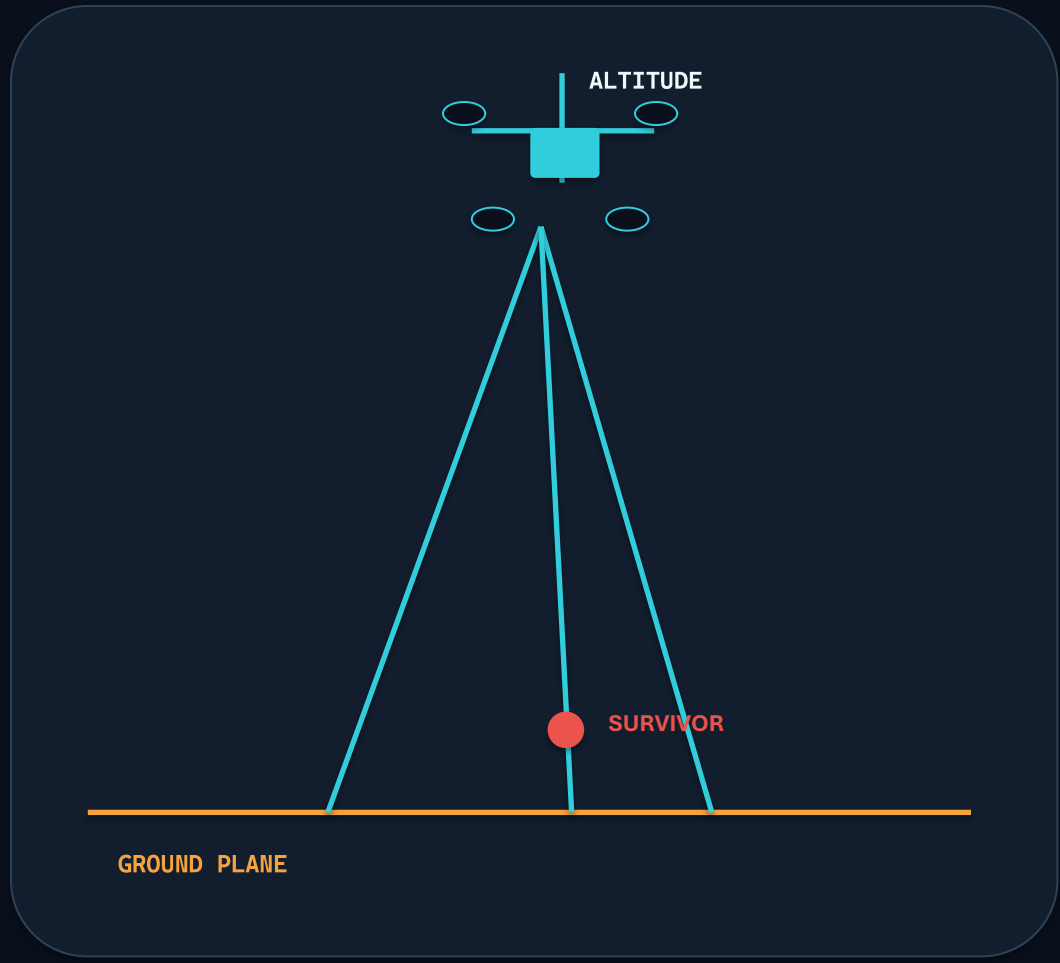
ARCHITECTURE SUPPORTS A FUTURE CUSTOM SURVIVOR DETECTOR • NO FINAL PHYSICAL ACCURACY CLAIM

Detection ≠ Location

A bounding box is not a latitude / longitude.

The system projects a camera ray through the detected pixel and intersects it with the ground model.

- Camera intrinsics + lens model
- Pixel + mounting position / rotation
- Drone position + altitude
- Roll, pitch, yaw / quaternion
- Frame timestamp + ground model



Pixel → World Coordinates

The transform chain is explicit, testable and quaternion-authoritative.



$$T_{world_camera} = T_{world_body} \times T_{body_camera}$$

pixel → camera ray → body ray → world ray → ground intersection → ENU → latitude / longitude

QUATERNION internal 3D orientation authority

Real Flight Disturbance Changes Geolocation

Wind is not a decorative animation.

Gazebo applies force. ArduPilot responds. The camera pose inherits the aircraft attitude.



CALM

Stable footprint; smaller pose variation.

CROSSWIND / GUST

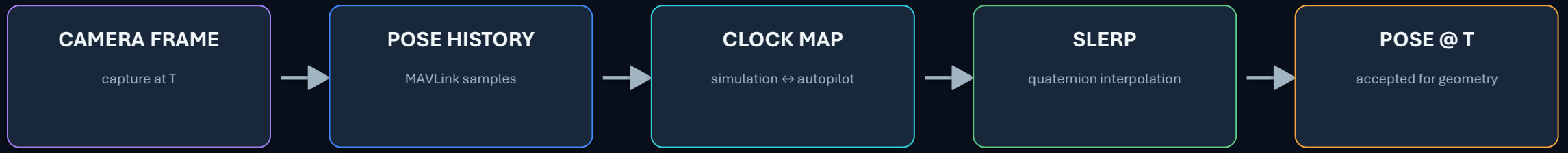
Natural attitude response shifts the projected ray.

PHYSICS STRESS TEST / EXTREME ~60° PEAK-TO-PEAK ATTITUDE VARIATION IS NOT REPRESENTATIVE FLIGHT VALIDATION

TIME SYNCHRONIZATION

The Right Pose at the Right Frame

Using the latest pose can be wrong during rapid motion.



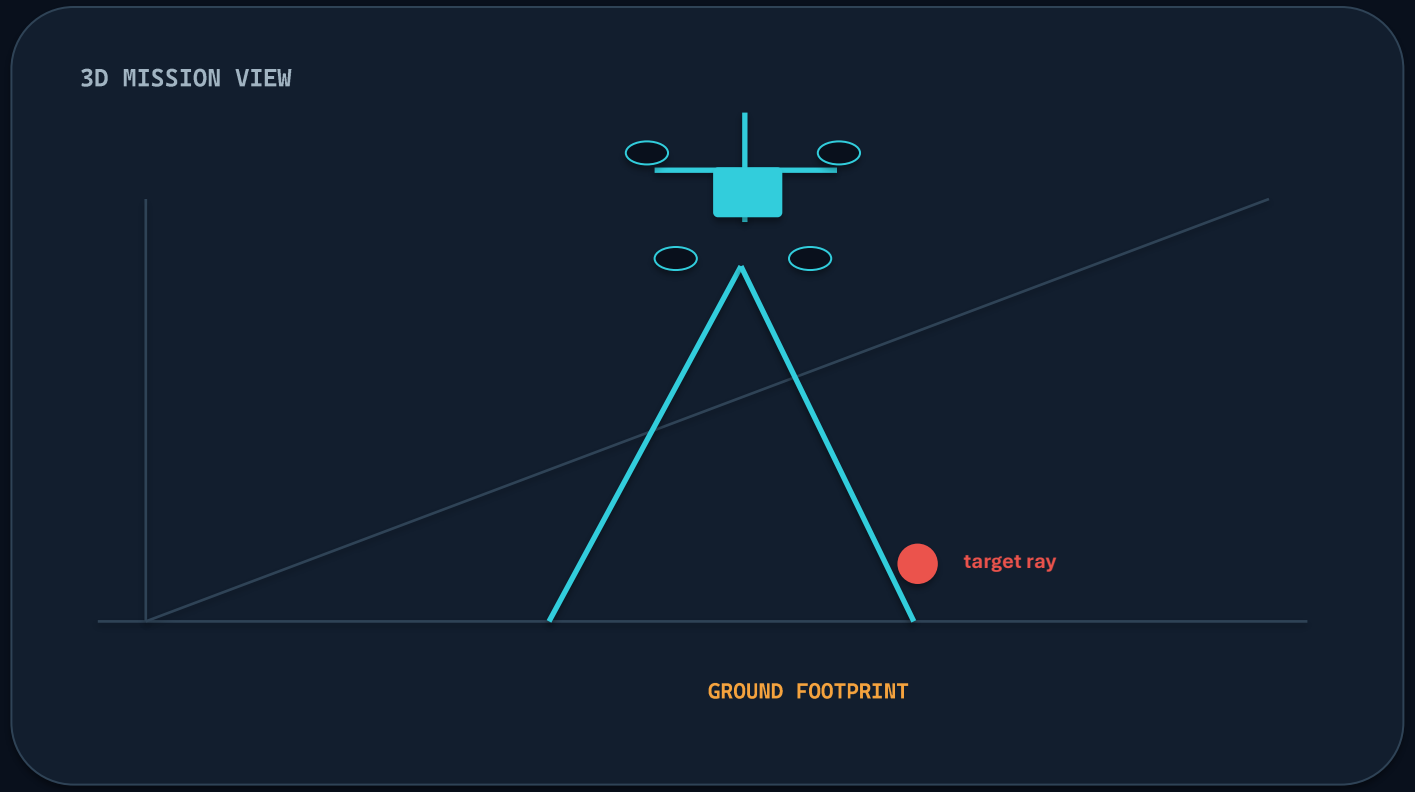
`unmapped capture timestamp → REJECT GEOLOCATION`

The current Phase-2 path fails closed instead of substituting frame receipt time.

HONEST safety-oriented engineering

Spatial Digital Twin

Two views of reality: autonomy estimate vs simulation ground truth.



VISUALIZES

Position • quaternion • paths • frustum • footprint

EXPLAINS

Target rays • uncertainty • sectors • deliveries

REPLAYS

Mission timeline • engineering inspector • observer

`0.000e+00 m / deterministic Python ↔ TypeScript spatial fixture discrepancy`

SIMULATION

High-Fidelity Closed-Loop Simulation Lab

A configurable digital test range for the full autonomy loop.



DRONE CONFIG

Frame • mass • payload • speed • inertia / drag

CAMERA CONFIG

Resolution • FPS • FOV • mount • latency / noise

ENVIRONMENT

Wind • gusts • lighting • origin • hidden targets

`Gazebo Harmonic / gz-sim 8.15.0 / official ardupilot_gazebo integration`

No Ground-Truth Leakage

The simulator knows the answer. Autonomy does not.

SIMULATOR KNOWS

- Actual survivor location
- True aircraft pose
- True wind
- Rendered visibility

TRUTH FIREWALL

AUTONOMY KNOWS ONLY

- Camera pixels
- MAVLink / EKF pose
- Detector output
- Mission state

Ground truth is used only after prediction for scoring / evaluation.

PIXEL-ONLY PERCEPTION

What the Virtual Drone Actually Sees

Rendered pixels first. Evaluation truth later.



Hidden 3D survivor → rendered camera → YOLO11n ONNX → bbox → confirmation → geolocation

SIMULATION DEV EVIDENCE

TP 63 • FP 3 • FN 438
Conditional precision 0.9545

BOUNDING BOX

P50 IoU 0.5762
P95 IoU 0.6879

QUALIFICATION STATUS

Phase 2 remains under qualification
frame-time chain incomplete

NOT FINAL REAL-WORLD DETECTOR ACCURACY • model SHA256: 417c59a7…6f2a5e2d • threshold 0.25

VERIFIED STATUS

Software Qualification

Software verified. Counts evolve as development progresses.

PRIMARY STAGE 12

328

328

LATEST PHASE 2

371

371 combined

INTENTIONAL SKIPS

9

9

SPATIAL FIXTURE

0.000e+00

0.000e+00 m

Python 335 passed / 9 skipped • Frontend 36 passed • Combined 371 passed / 9 skipped

OFFLINE

PASS • zero external runtime dependencies in qualified GCS

BUILD

Production frontend build PASS

DISPLAY QA

1366×768 • 1536×864 • 1920×1080 • 125% zoom

DETERMINISTIC REPLAY PASS • COMPETITION MODE RESTRICTIONS PASS • ZERO HORIZONTAL OVERFLOW

Multi-Drone Qualification in ArduPilot SITL

Competition-scale software rehearsal, not physical flight.

VEHICLES	DELIVERIES	COVERAGE	INTERVENTIONS	OUTCOME
2	4	99.7453%	0	SUCCESS
2	4	99.7453%	0	SUCCESS

4 synthetic survivor targets • fleet distance 669.67 m • both vehicles RTL / landed / disarmed

SCALE

99,584.34 m^2 ≈ 9.9584 ha boundary

REGISTRY

10-survivor registry PASS
3-vehicle smoke test PASS

RECOVERY

Duplicate prevention PASS
Vehicle-loss reallocation PASS

ARDUPILOT SITL / ArduCopter 4.8.0 DEV / commit af2a1baf…78d11a

One Operator — Complete Mission Visibility

A single local control surface for setup, supervision and review.

MISSION STATUS / FLEET HEALTH

T+ 08:42

2D / 3D MISSION MAP



FLEET / TARGET CARDS

- DRONE 01 SEARCH
- DRONE 02 DELIVER
- TARGET 04 CONFIRMED
- PAYLOAD 3 / 4

SETUP MISSION SIMULATION ENGINEERING REVIEW

Built for Safety and Infrastructure Independence

SAFETY BY DESIGN

Native ArduPilot RTL / LAND respected
Geofence integration
Emergency recall + safety abort
Health freshness + stale telemetry
Bounded queues + durable mission events
Recovery after service restart
Critical-failure target / drop gating

OFFLINE BY DESIGN

Local network only
No cloud inference
No GSM / LTE / 5G dependency
No public Wi-Fi
Locally bundled frontend assets
Local database
Offline runtime compliance verified

COMPETITION MODE / allowed controls: mission load, START, safety abort, emergency recall

PHYSICAL STATUS

Hardware Integration

The hardware path is defined. Physical validation is still pending.

AIRBORNE

Raspberry Pi 5 • OV9281 mono global shutter

FLIGHT

Pixhawk 2.4.8 • ArduCopter • M8N-class GNSS

GROUND

Raspberry Pi 4 • local GCS

PAYLOAD

Emergency kit mechanism • custom airframe

CURRENT STATUS

SOFTWARE	release candidate
SIMULATION	high-fidelity active
PHYSICAL	pending / diagnosis

- Battery packs / safe power
- OV9281 hardware troubleshooting
- Pixhawk / M8N validation
- Camera calibration + payload test
- Real flight validation

PENDING ≠ FAILED • no physical-flight or payload-delivery claim is made

VALIDATION ROADMAP

From Digital Twin to Physical Rescue Flight

Progression is gated by evidence, not by optimism.



COMPLETED — Core architecture • planning • single-drone autonomy • fleet SITL • GCS • 3D twin

ACTIVE — Pixel-only perception integration • high-fidelity Gazebo physics

NEXT — Frame-time mapping • sensor-noise realism • full simulated rescue mission

HARDWARE GATE — Camera calibration • physical geolocation • payload • real multi-drone flight

RescueSwarm is designed not merely to detect survivors — but to autonomously search, locate, coordinate, deliver aid and safely complete the mission.

CLOSING STATEMENT

RescueSwarm

Autonomous Intelligence for Faster Emergency Response

Search. Detect. Locate. Deliver. Return.

AUTONOMOUS | COLLABORATIVE | OFFLINE | POSE-AWARE | VISION-DRIVEN | SAFETY-ORIENTED

SOFTWARE VERIFIED • SITL / GAZEBO SIMULATION VERIFIED • PHYSICAL HARDWARE PENDING